

Multiple Choice

1. **Answer:** A

Options: A) 7.9 ohms; B) 3.9 ohms; C) 10.2 ohms; D) 6.3 ohms; E) 12.5 ohms

Note: Correct option: A - 7.9 ohms

2. **Answer:** C

Options: A) 190; B) 200; C) 170; D) 180; E) 120 *Note:* Correct option: C - 170

2. **Answer:** A

Options: A) 0.1256 A, 0.2 A, 0.1256 A; B) 0.1256 A, 0.3 A, 0.1256 A; C) 0.2345 A, 0.1 A, - 0.4567 A; D) 0.2 A, 0.1 A, 0.2 A; E) 0 A, 0.1 A, 0 A

Note: Correct option: A - 0.1256 A, 0.2 A, 0.1256 A

3. **Answer:** C

Options: A) $y(x) = c_1 * \sin(x) + c_2 * \cos(x)$; B) $y(x) = c_1 * y_1(x)^2 + c_2 * y_2(x)^2$; C) $y(x) = c_1 y_1(x) + c_2 y_2(x)$; D) $y(x) = c_1 * x * y_1(x) + c_2 / (x * y_2(x))$; E) $y(x) = c_1 * y_1(x) / (c_2 * y_2(x))$

Note: Correct option: C - $y(x) = c_1 y_1(x) + c_2 y_2(x)$

4. **Answer:** A

Options: A) gross error.; B) instrumental error.; C) systematic error.

Note: Correct option: A - gross error.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '- 5 mph'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $h[n] = 2 (1/3)^n u[n] - 3 (1/3)^n u[n]$ '.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 4.73. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 400 Mpa. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key